

FasterCap 3D Input Files

Dielectric definitions ('D' statement)

Syntax: D <file> <outperm> <inperm> <xoffset> <yoffset> <zoffset> <xref> <yref> <zref> [-]

The 'D' element at the beginning of a line defines a dielectric. The geometry of the dielectric surface is further specified in the file <file>. The dielectric surface is intended as the interface between two regions with relative permittivities <outperm> and <inperm>. The relative permittivities <outperm> and <inperm> can be complex valued, in the format $eRe - j eIm$, where eRe is the real part of the complex permittivity value, and eIm is the imaginary part.

The dielectric can be translated with respect to the coordinates defined in <file> by an offset ($xoffset$, $yoffset$, $zoffset$), thus allowing to reuse the same geometric definitions multiple times.

The reference point ($xref$, $yref$, $zref$) and the optional - argument are used to specify which side of the dielectric interface has which permittivity. More specifically, the reference point is assumed to lie on the <outperm> side of all the panels in <file>. The optional - argument indicates that ($xref$, $yref$, $zref$) instead lies on the <inperm> side. The reference point is not translated, i.e. the offset only applies to the elements specified in <file>.

Remark: it is the user's responsibility to make sure that the reference point is on the same side of the dielectric interface for **all** panels. That is, each panel is evaluated as stand-alone with respect to the reference point, to define its <outperm> and <inperm> sides. There is no concept of a external and internal side of a surface specified by a group of panels, even if specified in the same file with the same conductor names, since FasterCap maintains no topological information.

For complex shaped surfaces, FasterCap supports also the option to specify the reference point panel by panel, see [3D Triangular panel definitions \('T' statement\)](#) and [3D Quadrilateral panel definitions \('Q' statement\)](#) for details about the syntax. You can also mix the two type of panel definitions, i.e. panels without and panels with a per-panel reference point specification. In this case, panels without a reference point specified per panel will use the reference point specified for the whole dielectric interface in the 'D' statement.

Example 1:

```
D sphere.txt 1.0 2.0 0.0 0.0 0.0 0.0 0.0 0.0 -
```

This input file fragments specifies a dielectric interface whose geometry is defined in the file *sphere.txt*. The interior of the sphere is filled with a material with relative permittivity equal 2.0, since the reference point is centered on the origin and the optional - argument is specified; while the dielectric medium outside the sphere has a relative permittivity equal to 1.0.

Example 2:

```
D sphere.txt 1.0 3.0-j0.02 0.0 0.0 0.0 0.0 0.0 0.0 -
```

This input file fragments specifies a dielectric interface whose geometry is defined in the file *sphere.txt*. The interface is between air (relative permittivity equal to 1.0) and a lossy dielectric with complex relative permittivity equal to $3.0 - j0.02$ (e.g. with real part equal to 3.0 and imaginary part equal to 0.02).